Temperature

When fishes are exposed to high temperature their central nervous system recognizes the threat and gets activated. It shows physiological stress response and release catecholamines hormones whose secretion gets continuous increase with the increase of stress.[7] The secretion of cortisol hormones is not as rapid as release of catecholamines hormones. Fishes seem to tolerate small fluctuations in temperature.[8] The fish increases its metabolic activities with increase in temperature but only upto a threshold temperature. Higher temperatures increase the respiration rate, which increases the need of oxygen. It has effects on the rate of reproduction, usually decreasing it.

pH

Increase in pH up to 8.3 usually does not harm the fish productivity, but decreases it gradually. The optimum pH for fish's maximum productivity is between 6.5 to 7.4. Any increase or decrease in pH from the threshold can affect the hormonal equations, leading to loss in vigor or egg production.

Humidity

The humidity is not seen to exert any effects on the metabolic activity of the fish.

Turbidity

Initially turbidity doesnot seem to affect the fishes metabolic activity, however on the long run, it decreases the photosynthetic rate, hampering the phytoplankton and zooplankton's mass. This can lead to starvation in the absence of any addition of food manually.

Starvation

The physiological processes affect organs like liver, skeletal muscles, stomach, during starvation. Changes in the metabolic processes like breathing, oxidation states, intermediates formed, enzymatic action etc are all followed by starvation.

Initial starvation process is followed by lessened respiration rate, slow movement, and finally cessation of any bodily metabolism. Long periods of starvation usually accounts for hibernation. However in this case, repeated starvation caused loss in mass, less vigor, no reproduction and eventually death.

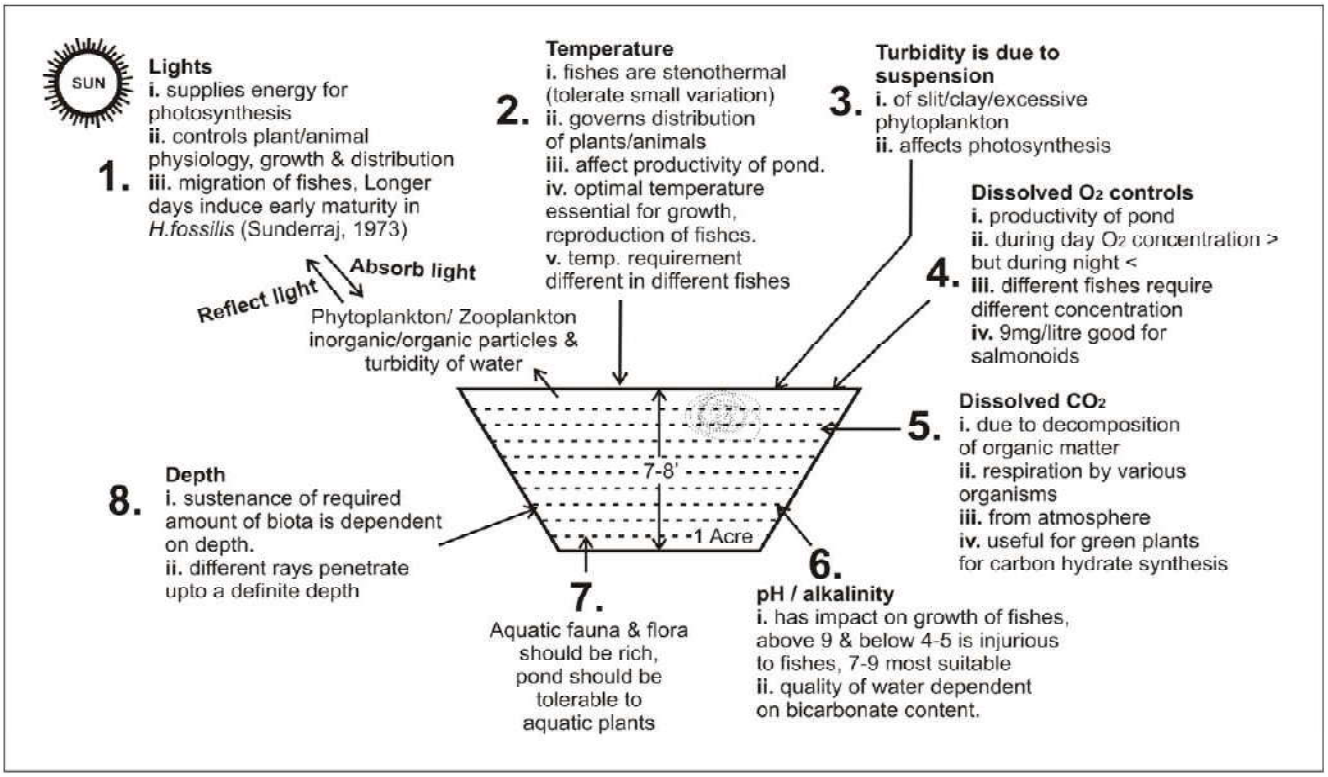


Fig. 1- Conditions of a pond is dependent upon above parameters

CONCLUSION

From the above study it was concluded that the stress can affect the production of fish. All types of stressors can cause depletion of glycogen level in muscle which adversely effect the quality. This study also concludes that fishes are good indicators of aquatic pollution. By maintaining the water quality properly one can produce a good quality of fishes.

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Survey report on National Research Centre for Orchids

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Abstract- National Research Centre for Orchids was established on 5th October, 1996 by the Indian Council of Agricultural Research (ICAR), New Delhi to organise programme on improvement in productivity quality and commercialization of orchids. Orchids are one of the high value crop cultivated for cut flower and pot plants in Sikkim. Usually, the orchids are grown on the plant in Sikkim. Orchids are one of the most sought after flower, they are expensive, elegant dainty, rare to find and possess a miraculous beauty of courses not everyone can afford to buy an orchid on their way home as a gift to often. However, even a one-time gift of an orchid signifies feelings of love romance and friendship in this project we will study history of orchids, findings of orchids, growth & conservation of orchids etc.

Key words: Orchids, NRCO, ICAR, Elegant, Miraculous, Conservation

INTRODUCTION

Orchids are mostly found in the North Easternmost tip of India, like Arunachal Pradesh, Sikkim Darjeeling, Ranipool, Kalimpong, Rangpo, Gangtok, Pakyong, Dikling, and adjoining areas. Sikkim is called 'The Land of Rising Sun' it is also fondly called 'Land of Down lit Mountains' or the 'Orchid State' for the abundance of a beautiful and rare orchid species found here. Eurasia, America and Canada are known as the home of orchids but today they are found almost in every part of the world. A French botanist Noel Bernard was the first to discover this magnificent flower, back in the year 1899, he came across seeds of a terrestrial orchids *Neottia*. He found that these seeds were germinated in some fruit which were infected by mould.

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Another discovery was by Charles Darwin, he noted a single orchid plant *Ophrys*. Charles Darwin, in a letter to Joseph Hooker, wrote, I never was more interested in any subject in my life than orchid.

Orchids constitute on order of royalty in the world of ornamental plants and they are of horticulture importance and play a very useful role in balance the forest ecosystem. Theophrastus, a student of Plato noted the round praised bulbous of one Common European Orchids and named it the Greek word for testicle. Later this name was applied not only to one genes but also to the entire Family Orchidaceae.

MATERIALS & METHODS

Stationary items, DSLR camera, Mobile Temperature G.P.S etc. NRCO at Dikling Road is a wonderful place where I met Director of NRCO, Dr. Rampal Sir, there I

visited laboratory, had taken photographs of Orchids of Poly houses and Net houses. In the NRCO the research area was divided in two parts the NRCO Office and Administrative Building, where researchers works in their lab. In this building library and conference hall are well maintained. On 5th October 1996, ICAR established NRCO on the basis of planning commission in 5 year plan institute works, as for vision for research & development activities orchids commercialization and sustainable utilization the library system.

OBSERVATION

Tissue culture lab were operating using multiplication of Orchids. Entomology section, Plant pathology section, Horticulture, Soil Science lab were well maintained. 325 different species of orchids were grown in 4 acre land for conservation purpose. Growing Orchids in polyhouse chamber, Tropical Orchid chamber, Natural Net House, is a very often process. Germination of polyhouse Rapidly Growth in Polyhouse. Rapidly increases Growth of Net house and Lower Part of the NRCO where cold climate are found Orchids are attached with the wood or in Natural condition.

DISCUSSION

There are four Veda in Sanskrit literature Rig Veda Sam Veda, Ayur Veda and Atharva Veda, written between 400 and 1600 B.C. Ayur Veda deals with use of plant drugs, it is considered as an Up Veda. The Indian orchids were brought to the notice of the world by Charak the great Indian medicine man. Charak the great Indian medicine man brings in notice of the world about the existence of orchid around 125 A.D. He described 'VANDA' and several others orchids in his book Charak Samhita which provides description of present known orchid like *Flickingeria, Malaxis, Eulophia*, species[1] having tremendous horticulture and medicinal value. It is believed that Chinese were the first to cultivate and describe orchids. They were certainly the first to describe orchids for medicinal uses.[2] In early 200 BC the Chinese Pharmacopeia "The San Nung Pen Tsao Ching" mention Dendrobium as a stringent, analgesic, anti-inflammatory substances. Reinikka reports a Chinese legend 'Shunnug' who described *Striata* and a *Dendrobium* species in his book 'Materia Medica' of the 28th Century B.C. One of the earliest nomenclature used around 937-286 BC for Orchids was that:- "The Orchids a generic name which stands for the testicles like things are basically the pair of underground tuber of many European terrestrial orchids".

In breeding programme, selection of good and healthy plant and flower by visual observation accounts to a great extent. Evidences of natural hybridization. Occurring among wild species were reported[3] in Dendrobium, Offspring's of reciprocal crosses show variations in characters like cane length and flower colour, flower size, flowering season and flower yield.[4] Selection of flower size, and flower colour are effective in Dendrobium improvement programme.[5] In *Cymbidium*, fragrance is the most important character sought after by breeders. *Cymbidium munronianum* has been used as parent in several breeding programme.[6] The higher order hybrid, Paphiopedilum F.C Puddle with six species in parentage plays a predominant role in breeding for white flower colour in the genus.[7] An extensive breeding programme in phalaenopsis using 29 wild species and 873 varieties were studied by a group of workers[8] for development of new hybrids inter:- group hybridization between Phalaenopsis type Dendrobium extends flowering season, expands the range of flower colour and shapes and increases the flowering.[9] Majority of commercially grown orchids today are hybrids derived from Arachnis, Vanda, Renanthera, Ascocentrum, Cymbidium, Catteleya, Dendrobium, Oncidium, Phalaenopsis and Paphiopedilum.[10] *Vanda sanderiana* and *Vanda coerulea* are the two important Vandal species found in the back ground of most of the Vandaceous hybrids.[11] Most orchids have two basic sets (diploid, 2x) of chromosomes. Tetraploid plants are more Fertile bearing flowers of better texture, bigger and more intense colouration.[12] Many Indian species have earned worldwide recognition in breeding program due to their inherent attractiveness coupled with their ability to transmit these character to hybrids. Some of the leading species are *Aerides multiflorum, Cymbidium devonianum, C.lourianum, C.tracyanum, C.elegans, Dendrobium aggregatum, Dendrobium chrysotoxum, Dendrobium formosum, Dendrobium nobile, Paphiopedilum venustum, Vanda coerulea* etc.[13]

CONCLUSION

Collection of orchids from their natural bodies should be made illegal unless the same is made for scientific study or for propagation and preservation at